

Project Design Phase-II

Solution Requirements (Functional & Non-Functional)

Parameter	Details
Date	12 February 2026
Team ID	LTVIP2026TMIDS79181
Project Name	Smart Sorting – Rotten Fruit Detection using Machine Learning
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Image Input & Validation	Accept fruit image uploads and validate file format and size.
FR-2	Fruit Classification	Classify fruit image as Fresh or Rotten using the machine learning model.
FR-3	Image Preprocessing	Perform preprocessing steps such as image resizing, normalization, and augmentation.
FR-4	Model Inference	Load trained fruit classification model and return prediction result.
FR-5	Prediction Output	Display prediction results clearly in the web interface.
FR-6	User Interface	Provide a simple interface for users to upload fruit images and view results.
FR-7	Dataset & Model Management (Admin)	Allow model retraining or replacement when new fruit datasets are available.

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a simple and intuitive interface for users to upload fruit images and view classification results.
NFR-2	Security	Ensure secure handling of uploaded images and perform strict input validation.
NFR-3	Reliability	Maintain consistent preprocessing and prediction using saved model artifacts.
NFR-4	Performance	Deliver fast fruit classification with prediction time less than 1 second per image .
NFR-5	Availability	System should be accessible whenever users need to test fruit images.
NFR-6	Scalability	Support larger datasets and extended fruit detection capabilities.
NFR-7	Maintainability	Use modular and well-documented code to allow easy updates and model retraining.
NFR-8	Portability	Support deployment on Windows/Linux systems and allow future cloud deployment.